Demostrar el despliegue con Docker

1. Mostrar las imágenes Docker creadas

Comando:

```
docker images
```

```
PS C:\Users\Diego\Desktop\TFG> docker images
                                                    i Info →   U   In Use
IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
diegolarra06/adopta-backend:1.0      e22d0b013c02       527MB         157MB       U
diegolarra06/adopta-db:1.0           08be41f423a3       1.1GB         238MB       U
diegolarra06/adopta-frontend:1.0
```

Muestra las imágenes Docker que se han construido para la aplicación. En mi caso he creado tres imágenes: una para la base de datos MySQL, otra para el backend Spring Boot y otra para el frontend Vue con Nginx.

2. Mostrar la red Docker creada

Comando:

```
docker network ls
```

```
PS C:\Users\Diego\Desktop\TFG> docker network ls
NETWORK ID     NAME         DRIVER    SCOPE
e1ea9421b516   adopta-red   bridge    local
2907cf47a985   bridge       bridge    local
```

He creado una red propia llamada `adopta-red` para que los contenedores puedan comunicarse entre ellos por nombre. Por ejemplo, el backend se conecta a la base de datos usando el nombre del contenedor `mysql-adopta`.

3. Mostrar los volúmenes creados

Comando:

```
docker volume ls
```

```
PS C:\Users\Diego\Desktop\TFG> docker volume ls
DRIVER    VOLUME NAME
local     adopta_mysql_data
local     adopta_uploads
local     volumen1
```

He creado dos volúmenes: uno para guardar los datos de MySQL y otro para guardar los archivos subidos por la aplicación. Así, aunque se borren los contenedores, los datos importantes pueden mantenerse.

4. Arrancar la base de datos

Si el contenedor ya existe, puedes arrancarlo con:

```
docker run -d --name mysql-adopta --network adopta-red -p 3307:3306
-v adopta_mysql_data:/var/lib/mysql -e
MYSQL_ROOT_PASSWORD=PracticaRoot diegolarra06/adopta-db:entrega-
final
```

Este comando arranca el contenedor de MySQL.

La opción `--name mysql-adopta` le da un nombre al contenedor.

La opción `--network adopta-red` lo conecta a la red común.

La opción `-p 3307:3306` publica MySQL en el puerto 3307 del ordenador porque el puerto 3306 ya estaba ocupado.

La opción `-v adopta_mysql_data:/var/lib/mysql` guarda los datos de MySQL en un volumen.

La variable `MYSQL_ROOT_PASSWORD=PracticaRoot` establece la contraseña de root de MySQL.

5. Comprobar logs de MySQL

Comando:

```
docker logs mysql-adopta
```

```
2026-05-09T15:30:39.270877Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
2026-05-09T15:30:39.540623Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
2026-05-09T15:30:39.540671Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported
2026-05-09T15:30:39.543170Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is acc
2026-05-09T15:30:39.562868Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Socket: /var/run/mysqld/mysqlx.sock
2026-05-09T15:30:39.563027Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.4.9'  socket: '/var/run/mysqld/m
2026-05-09 15:30:39+00:00 [Note] [Entrypoint]: Temporary server started.
'/var/lib/mysql/mysql.sock' -> '/var/run/mysqld/mysqld.sock'
Warning: Unable to load '/usr/share/zoneinfo/iso3166.tab' as time zone. Skipping it.
Warning: Unable to load '/usr/share/zoneinfo/leap-seconds.list' as time zone. Skipping it.
Warning: Unable to load '/usr/share/zoneinfo/leapseconds' as time zone. Skipping it.
Warning: Unable to load '/usr/share/zoneinfo/tzdata.zi' as time zone. Skipping it.
Warning: Unable to load '/usr/share/zoneinfo/zone.tab' as time zone. Skipping it.
Warning: Unable to load '/usr/share/zoneinfo/zone1970.tab' as time zone. Skipping it.
2026-05-09 15:30:41+00:00 [Note] [Entrypoint]: Creating database adopta_un_companero

2026-05-09 15:30:41+00:00 [Note] [Entrypoint]: /usr/local/bin/docker-entrypoint.sh: running /docker-entrypoint-initdb.d/01_init.sql


2026-05-09 15:30:41+00:00 [Note] [Entrypoint]: Stopping temporary server
2026-05-09T15:30:41.676840Z 12 [System] [MY-013172] [Server] Received SHUTDOWN from user root. Shutting down mysqld (Version: 8.4.9).
2026-05-09T15:30:42.908785Z 0 [System] [MY-010910] [Server] /usr/sbin/mysqld: Shutdown complete (mysqld 8.4.9)  MySQL Community Server - GPL.
2026-05-09T15:30:42.908820Z 0 [System] [MY-015016] [Server] MySQL Server - end.
2026-05-09 15:30:43+00:00 [Note] [Entrypoint]: Temporary server stopped

2026-05-09 15:30:43+00:00 [Note] [Entrypoint]: MySQL init process done. Ready for start up.

2026-05-09T15:30:43.695712Z 0 [System] [MY-015015] [Server] MySQL Server - start.
2026-05-09T15:30:43.900297Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.4.9) starting as process 1
2026-05-09T15:30:43.904726Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
2026-05-09T15:30:44.111153Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
2026-05-09T15:30:44.331024Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
2026-05-09T15:30:44.331063Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported
2026-05-09T15:30:44.333517Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is acc
2026-05-09T15:30:44.360923Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysq
2026-05-09T15:30:44.361149Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.4.9'  socket: '/var/run/mysqld/m
```

Se comprueba que la base de datos ha arrancado correctamente y que se ha importado el script SQL inicial.

6. Arrancar el backend

Si el contenedor ya existe:

```
docker start backend-adopta
```

Comando completo usado para crearlo:

docker run -d --name backend-adopta --network adopta-red -p 8080:8080 -v adopta_uploads:/app/uploads -e SPRING_DATASOURCE_URL="jdbc:mysql://mysql-adopta:3306/adopta_un_companero?useSSL=false&serverTimezone=UTC&characterEncoding=UTF-8&allowPublicKeyRetrieval=true" -e SPRING_DATASOURCE_USERNAME=root -e SPRING_DATASOURCE_PASSWORD=PracticaRoot diegolarra06/adopta-backend:entrega-final

Este comando arranca el backend desarrollado con Spring Boot.

La opción --network adopta-red permite que el backend pueda conectarse a MySQL.

La conexión a la base de datos se pasa con la variable SPRING_DATASOURCE_URL.

En vez de usar localhost, se usa mysql-adopta, que es el nombre del contenedor de la base de datos dentro de la red Docker.

El puerto 8080:8080 permite acceder al backend desde el ordenador en:

http://localhost:8080

7. Comprobar logs del backend

Comando:

docker logs backend-adopta

```
PS C:\Users\Diego\Desktop\TFG> docker logs backend-adopta

  .   ____          _            __ _ _
 /\\ / ___'_ __ _ _(_)_ __  __ _ \ \ \ \
( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/

 :: Spring Boot ::                (v3.3.5)

2026-05-09T15:31:01.766Z  INFO 1 --- [AdoptaUnCompaneroApplication] [           main] c.d.a.AdoptaUnCompaneroApplication      : Starting AdoptaUnCompaneroApplication v0.0.1-SNAPSHOT using Java 17.0.19 with PID 1 (/app/
app.jar started by root in /app)
2026-05-09T15:31:01.770Z  INFO 1 --- [AdoptaUnCompaneroApplication] [           main] c.d.a.AdoptaUnCompaneroApplication      : No active profile set, falling back to 1 default profile: "default"
2026-05-09T15:31:02.937Z  INFO 1 --- [AdoptaUnCompaneroApplication] [           main] .s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT mode.
2026-05-09T15:31:03.091Z  INFO 1 --- [AdoptaUnCompaneroApplication] [           main] .s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in 139 ms. Found 11 JPA repository interfaces.
2026-05-09T15:31:04.286Z  INFO 1 --- [AdoptaUnCompaneroApplication] [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat initialized with port 8080 (http)
2026-05-09T15:31:04.308Z  INFO 1 --- [AdoptaUnCompaneroApplication] [           main] o.apache.catalina.core.StandardService   : Starting service [Tomcat]
2026-05-09T15:31:04.309Z  INFO 1 --- [AdoptaUnCompaneroApplication] [           main] o.apache.catalina.core.StandardEngine    : Starting Servlet engine: [Apache Tomcat/10.1.31]
2026-05-09T15:31:04.390Z  INFO 1 --- [AdoptaUnCompaneroApplication] [           main] o.a.c.c.C.[Tomcat].[localhost].[/]       : Initializing Spring embedded WebApplicationContext
2026-05-09T15:31:04.393Z  INFO 1 --- [AdoptaUnCompaneroApplication] [           main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 2553 ms
2026-05-09T15:31:04.698Z  INFO 1 --- [AdoptaUnCompaneroApplication] [           main] com.zaxxer.hikari.HikariDataSource       : HikariPool-1 - Starting...
2026-05-09T15:31:04.951Z  INFO 1 --- [AdoptaUnCompaneroApplication] [           main] com.zaxxer.hikari.pool.HikariPool        : HikariPool-1 - Added connection com.mysql.cj.jdbc.ConnectionImpl@93501be
2026-05-09T15:31:04.954Z  INFO 1 --- [AdoptaUnCompaneroApplication] [           main] com.zaxxer.hikari.HikariDataSource       : HikariPool-1 - Start completed.
2026-05-09T15:31:05.003Z  INFO 1 --- [AdoptaUnCompaneroApplication] [           main] o.hibernate.jpa.internal.util.LogHelper  : HHH000204: Processing PersistenceUnitInfo [name: default]
2026-05-09T15:31:05.100Z  INFO 1 --- [AdoptaUnCompaneroApplication] [           main] org.hibernate.Version                    : HHH000412: Hibernate ORM core version 6.5.3.Final
2026-05-09T15:31:05.142Z  INFO 1 --- [AdoptaUnCompaneroApplication] [           main] o.h.c.internal.RegionFactoryInitiator    : HHH000026: Second-level cache disabled
2026-05-09T15:31:05.564Z  INFO 1 --- [AdoptaUnCompaneroApplication] [           main] o.s.o.j.p.SpringPersistenceUnitInfo      : No LoadTimeWeaver setup: ignoring JPA class transformer
```

permite comprobar que el backend ha arrancado y que ha conectado correctamente con la base de datos.

8. Arrancar el frontend

Si el contenedor ya existe:

```
docker start frontend-adopta
```

Comando completo usado para crearlo:

```
docker run -d --name frontend-adopta --network adopta-red -p 80:80
diegolarra06/adopta-frontend:entrega-final
```

Explicación: Este comando arranca el frontend de la aplicación. El frontend está construido con Vue y se sirve mediante Nginx.

9. Comprobar que los tres contenedores están activos

Comando:

docker ps

Explicación: muestra los contenedores que están actualmente en ejecución. Deben aparecer los tres contenedores principales de la aplicación.

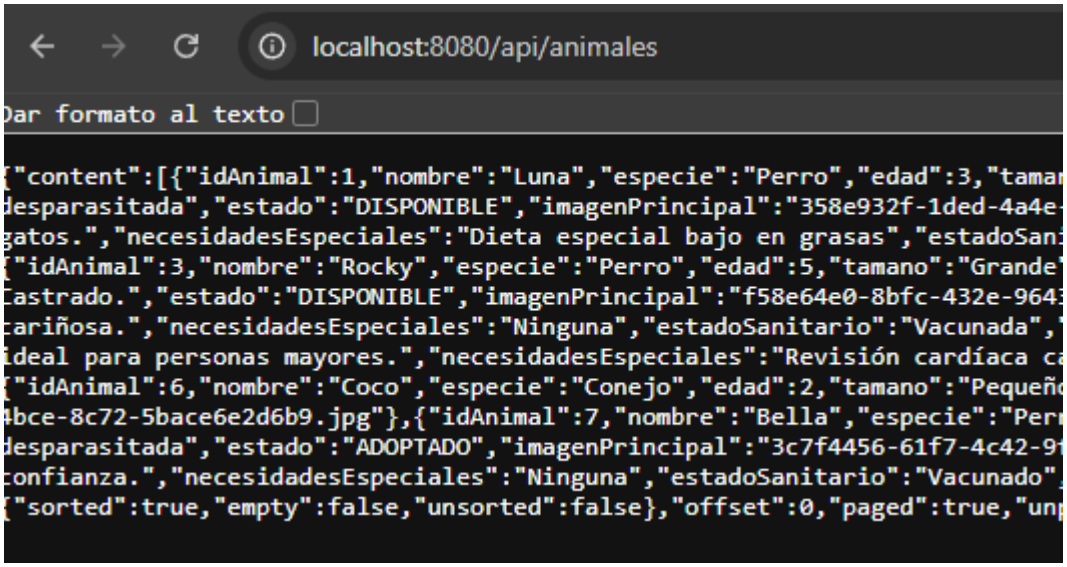
```
PS C:\Users\Diego\Desktop\TFG> docker ps
CONTAINER ID   IMAGE                        COMMAND                  CREATED        STATUS        PORTS                                          NAMES
17dbaa0a0ff4   diegolarra06/adopta-backend:1.0   "java -jar app.jar"      18 minutes ago   Up 18 minutes   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp   backend-adopta
b55b14f34810   diegolarra06/adopta-db:1.0       "docker-entrypoint.s…"   18 minutes ago   Up 18 minutes   0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp   mysql-adopta
ab2fdbc42d86   diegolarra06/adopta-frontend:1.0 "/docker-entrypoint.…"   4 hours ago      Up 4 hours      0.0.0.0:80->80/tcp, [::]:80->80/tcp           frontend-adopta
PS C:\Users\Diego\Desktop\TFG>
```

10. Abrir la aplicación en el navegador

11. Comprobar que el frontend se comunica con el backend

1. Busca peticiones a:

http://localhost:8080/api/animales



Aquí se comprueba que el frontend está llamando correctamente al backend. Las peticiones deben devolver estado `200`, lo que significa que la comunicación entre frontend, backend y base de datos funciona correctamente.

12. Comandos para parar la aplicación

Comando:

docker stop frontend-adopta backend-adopta mysql-adopta

detiene los tres contenedores de la aplicación. Se detiene primero el frontend, luego el backend y finalmente la base de datos.

13. Comandos para volver a abrir la aplicación

Comandos:

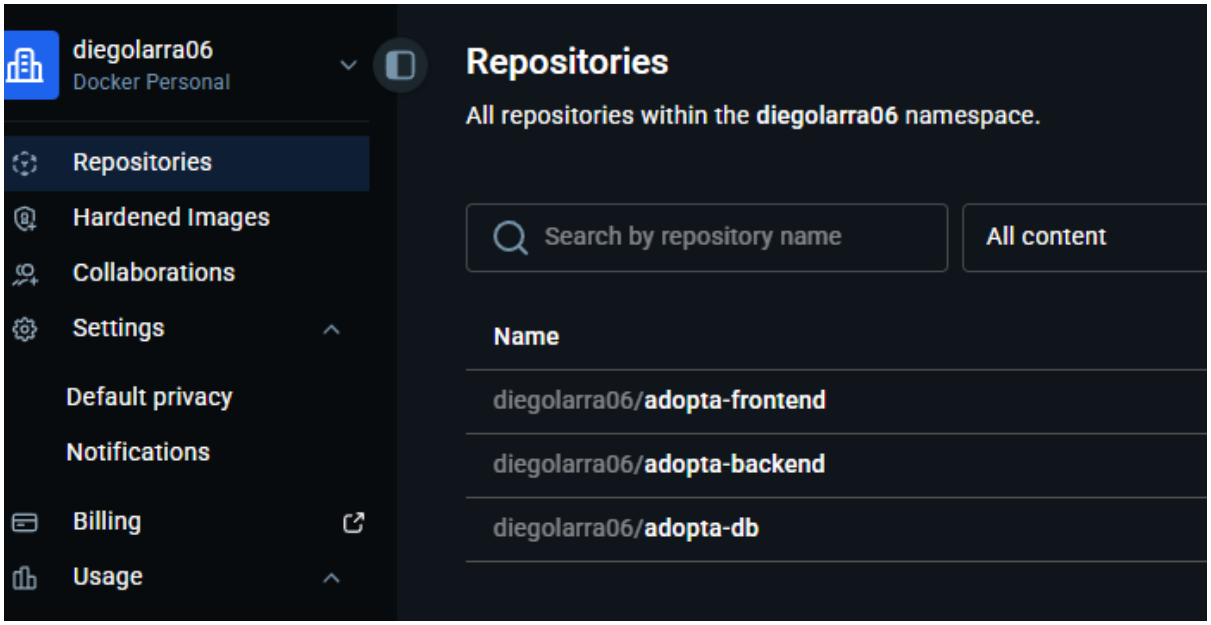
docker start mysql-adopta

docker start backend-adopta

docker start frontend-adopta

```
PS C:\Users\Diego\Desktop\TFG> docker start mysql-adopta
mysql-adopta
PS C:\Users\Diego\Desktop\TFG> docker start backend-adopta
backend-adopta
PS C:\Users\Diego\Desktop\TFG> docker start frontend-adopta
frontend-adopta
PS C:\Users\Diego\Desktop\TFG> docker ps
```

Si los contenedores ya existen, no es necesario volver a crearlos. Basta con arrancarlos de nuevo en orden: primero la base de datos, después el backend y finalmente el frontend.

14. Subir a DockerHub

```
PS C:\Users\Diego\Desktop\TFG> docker push diegolarra06/adopta-db:entrega-final
The push refers to repository [docker.io/diegolarra06/adopta-db]
56b4ed70176d: Pushed
1ba345b2c7a5: Pushed
c14e3f115d16: Pushed
3221f853f73e: Pushed
68c4d53bd503: Pushed
fc828cf283e3: Pushed
6405a7c8d60b: Pushed
35ff074259fa: Pushed
edf85873f64e: Pushed
0d744359c8b9: Pushed
344c14a42aea: Pushed
a85f525ae670: Pushed
entrega-final: digest: sha256:08be41f423a30995e20112c165f3c4019f4627e0e336a7bf2a640d5fe1e31a33 size: 856
PS C:\Users\Diego\Desktop\TFG> docker push diegolarra06/adopta-backend:entrega-final
The push refers to repository [docker.io/diegolarra06/adopta-backend]
c7bfdab0a425: Pushed
```